TABLE 153-2 Major Pathologic Differential Diagnosis of Sarcoidosis on Skin Biopsy

- Tuberculosis
- Atypical mycobacteriosis
- Fungal infection
- Reaction to foreign bodies: Beryllium, zirconium, tattooing, paraffin, etc.
- Rheumatoid nodules

A clinically important aspect of the pathology of sarcoidosis involves the development of fibrosis. Dense bands of fibroblasts may encase the ball-like granulomas. This fibrotic response can produce tissue destruction and organ dysfunction that is often irreversible. Currently available chemotherapeutic agents for sarcoidosis can effectively treat the granulomatous inflammatory response but not the fibrotic reaction.

DIAGNOSIS

The diagnosis of sarcoidosis requires a compatible clinical picture, histologic demonstration of noncaseating granulomas, and exclusion of other diseases capable of producing similar histology or clinical features. Because sarcoidosis is a diagnosis of exclusion, the diagnosis can never be confirmed with 100 percent certainty.

Normally, the diagnosis of sarcoidosis requires: - A compatible clinical picture - Histologic demonstration of noncaseating granulomas - Exclusion of other diseases with similar histology or clinical features

Alternatively, the diagnosis can be assumed without biopsy when the clinical presentation is typical and no other cause is evident.

The presence of noncaseating granulomas in a single organ does not conclusively establish sarcoidosis, as sarcoidosis is, by definition, a systemic disease involving multiple organs. Isolated skin granulomas should not be assumed to represent sarcoidosis, and alternative diagnoses must be excluded (see Table 153-2). Although a confirmed diagnosis of sarcoidosis requires granulomatous involvement in at least two separate organs, histologic confirmation is usually not required in the second organ.

Certain clinical presentations are highly specific for sarcoidosis—such as **Löfgren syndrome**, **Heerfordt syndrome**, and **asymptomatic bilateral hilar adenopathy**—and may allow diagnosis without biopsy.

Asymptomatic Bilateral Hilar Adenopathy

Asymptomatic bilateral hilar adenopathy on chest radiograph is highly suggestive of sarcoidosis. In asymptomatic patients with this finding, histologic confirmation may not be necessary if: - Physical examination is normal - Complete blood count and routine blood tests are normal - There is no prior history of malignancy

In such cases, serial chest radiographs should be monitored every 3 to 6 months. A biopsy should be performed if there is significant change in the radiographic appearance.

Panda and Lambda Sign on Gallium-67 Scan

On gallium-67 scanning: - **Panda sign**: Bilateral lacrimal and parotid gland uptake - **Lambda sign**: Bilateral hilar and right paratracheal uptake

These findings are highly specific for sarcoidosis and may eliminate the need for invasive diagnostic procedures. However, both signs are present in only a small percentage of patients.

NATURAL HISTORY AND PROGNOSIS

The granulomatous inflammation in sarcoidosis may remit spontaneously or with treatment. Overall, the prognosis is good.

- Pulmonary sarcoidosis resolves, improves, or stabilizes in **60 to 90 percent** of patients, even without therapy.
- Remissions commonly occur within the first 6 months after diagnosis, though resolution may take **2 to 5 years**.
- Prognosis is generally favorable for liver and peripheral lymph node involvement.
- Skin lesions may resolve with or without scarring or pigmentary changes.

Fibrosis is responsible for nearly all severe organ impairment in sarcoidosis and often leads to irreversible damage.